Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/06/2005
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Project Title:	Tissue-Specific Acute And Late Molecular Surveillance Of Particle Radiation Effects		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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	The detection of particle radiation induced alterations in the genome in different tissues in vivo is important for the assessment of both short- and long-term risks involved in space explorations. However, such studies are difficult due to (a) the lack of adequate selectable somatic target genes in mammalian systems and (b) the clonogenic potential of the somatic cells chosen. Mutagenic changes that affect the function of non-dividing tissues may be especially deleterious during long-term space flights. The advent of transgenic rodent gene mutation models has provided a means to rapidly assay, in a statistically reliable way, for mutations in the DNA from any tissue of the body over the period of the life span of the animal. These models are the only systems currently available for evaluating both short and long-term tissue specific mutagenic events. They have been accepted by regulatory agencies and are used in genetic toxicology studies on a wide variety of clastogenic and mutagenic agents. In the work proposed here, we have chosen to use a unique		

Task Description:	plasmid-based lacZ transgenic mouse system to determine the mutagenic potential of charged particle radiations relevant to the space environment. The lacZ transgenic mouse has been constructed so that every cell of the animal contains multiple copies of an integrated target gene. The use of this system allows us to recover and quantify radiation-induced mutations in all tissues of the animal, especially in tissues with limited cell turnover potential, and therefore allows tissue-specific radiation-induced changes to be measured as a function of time post radiation. The nature of these mutations can also be revealed by restriction fragment length polymorphisms (RFLP) and sequence analysis. We hypothesize that physical parameters of HZE radiation induces tissue-specific responses and aim to use the transgenic animal model in a multi-prong approach to extend our current knowledge to test three specific aims: (1) Examine the changes in transgene MF as a function of fluence, dose protraction of either pure or mixed particle beams. Beam characteristics in space are critical variables in space and tissue-specific biological effects are important in determining relative risks in space explorations; (2) Measure the dose- and temporal- dependence of HZE radiation-induced MF in the transgene in liver tissue and compare the results with those previously measured in the brain and the spleen. Excess relative risk in liver cancer and non-cancer liver disease or cirrhosis has been demonstrated in Atomic bomb survivors and thorotrast patients; (3) Advance our understanding of the molecular basis for mutation by surveying the transcription activation of target genes in the brain tissue. We will focus our research on examining the expression of proinflammatory cytokines, TNFalpha and interleukin-1, and the neuronal cell adhesion molecule NCAM-1. These genes have been shown to be induced in the mouse brain after low LET exposures. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The space atmosphere is comprised of particle radiation in a wide spectrum of higly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in lng-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Task Progress:	The study of genotoxic and mutagenic potential of particle beams in vivo is important to evaluating risks for future manned space missions. Using the transgenic mouse mutation model system, we are able to quantitate radiation-induced damage and recovery in tissues and the hematopoietic system relative to changes in beam quality. We participated in two NSRL campaigns during this period of performance. C57/Bl6/lacZ transgenic mice were exposed to graded doses of 600 MeV/u Iron ions during NSRL4 (Fall 2004), and 1 GeV/u Oxygen ions or 600MeV/u Silicon ions during NSRL-5 (Spring 2005). For each of these experiments, flow cytometric technique was used to map the changes in the frequency of MN-RET (fMN-RET) in peripheral blood samples in an early time course after exposure. Brain and spleen tissues, two tissues with contrasting proliferative capacities, were harvested at 2 to 4 months after exposure to evaluate mutant frequencies (MF) in the reporter transgene that is integrated into the genome of every cell of the organism. We noted a dramatic drop in the frequency of total circulating reticulocytes (fRET) shortly after each radiation exposure, a definitive indicator that particle radiation induced an overall dose-dependent systemic toxicity in the hematopoietic system. This was followed by complete recovery of RET counts within 1 week after exposure. The dose-dependent increase in the level of fMN-RET was observed within 24h after Silicon or Oxygen irradiations, but not after 600MeV/u Iron ions. Since the earliest damaged cells to enter the peripheral blood compartment are the cells that were in G2/M at the time of radiation exposure, differences in the fMN-RET at this early time may be used as an early assessment of biological effectiveness of each particle beam in the haematopoietic stem cell population. MN-RET levels peaked at 48h after exposure for all the beams examined followed by complete recovery at 1 week, independent of the LET of the radiation beam. Analysis of the profile of changes in fMN-RET as a function of incident particle fluence showed that particle beams with similar LETs, such as 290 MeV/u Carbon ions and 600 MeV/u Silicon ions, induce similar levels of increase in fMN-RET at about 48h after irradiation. We analyzed mutant frequencies in the brain tissues harvested at 8 weeks after 1 GeV/u Titanium ions irradiation. Results show that spontaneous lacZ MF in the control animals are consistent with those collected previously from other studies. We noted an increase in MF of up to 2-fold above spontaneous levels, as a function of dose after Titanium ion exposure. Comparison of these results with those previously reported for the 1 GeV/u Iron ions suggest that Titanium is less mutagenic than iron ions. As previously reported, lacZ MF in spleen after Titanium ions was significantly lower than Iron ion. Taken together, our results show that the yield of mutations in the reporter transgene in tissues is LET-dependent. Animals were exposed to 5 daily low dose (2 – 20 cGy) fractions of 1GeV/n Iron ions to simulate protracted radiation exposures in the space environment. We have previously reported that, using the peripheral blood fMN-RET as an early indicator for cytogenetic damage, fractionated exposures showed a dose sparing effect. Tissues from these animals were harvested at 8 and 16 weeks post irradiation. Mutations in the transgene in 8-week spleen tissues revealed that, at total doses =0.5 Gy, there were no differences between the two dosing regime. However, fractionated delivery of Iron particle radiation induced significant sparing effects in the spleen at higher doses. In the brain tissue, no significant sparing effects were observed up to 1 Gy, with significant sparing effects observed at 2 Gy. These results imply that the effect of protracted particle radiation exposure may be tissue-specific. We have demonstrated that the lacZ transgenic mouse mutation model is a suitable in vivo system to evaluate the cytogenetic damage in the haematopoietic system and the mutagenic potential of particle radiation effects in tissues. Our results to date suggest that early radiation responses of red cells are dependent on radiation quality. Such responses are temporally regulated and can be modulated by the dosing regime. Mutation analysis of two different tissues in the same animal highlights the importance of tissue specificity considerations in risk analysis. Protracted exposures of iron particle radiation induced an overall sparing effect in cytogenetic toxicity but the mutagenic potential of HZE particle radiation in the transgene after such exposures is highly tissue-dependent. The implications of these results may be important for risk estimation for manned space missions.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Articles in Peer-reviewed Journals	Chang PY, Bakke J, Orduna J, Lin S, Doppalaudi R. "Proton-induced genetic damage in lacZ transgenic mice." Radiat Res. 2005 Oct;164(4 Pt 2):481-6. PMID: 16187753 , Oct-2005

Presentation	Chang ,P.Y. , Bakke,J., Brune, J., Orduna, J., Doppalapudi, R. and Lin, S. "IMPACT OF TRP53 GENETIC BACKGROUND, RADIATION QUALITY AND DOSING REGIME ON PARTICLE RADIATION-INDUCED GENETIC DAMAGE IN VIVO" Bioastronautics Investigators' Workshop, Moody Gardens Hotel, Galveston, Texas Jan-2005
Presentation	Chang, P.Y.; Bakke, J.; Brune, J.; Orduna, J.; and Lin, S. "TISSUE-SPECIFIC HEAVY PARTICLE INDUCED EFFECTS IN TRANSGENIC ANIMALS" 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York. May-2005